IBM BOOTCAMP PROJECT REPORT

PhishGuard AI

Intelligent Phishing Detection System

"Detect Deception. Browse with Confidence."

IBM Bootcamp Project Report

Submitted in partial fulfillment of the IBM Bootcamp requirements

Technology Stack: Python • Streamlit • Scikit-learn • Random Forest

Domain: Cybersecurity • Machine Learning • Threat Intelligence

ZERO-EGRESS • OFFLINE • EXPLAINABLE AI

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1. Introduction

Phishing remains one of the most prevalent and damaging cyberattack vectors in the modern digital landscape. According to the Anti-Phishing Working Group (APWG), over 5 million phishing attacks were reported in 2023 alone, targeting individuals, enterprises, and government institutions. Attackers craft deceptive URLs that mimic trusted brands to harvest credentials, financial data, and personal information.

PhishGuard AI is a web-based cybersecurity application that leverages Machine Learning and heuristic analysis to detect phishing threats from URL structure alone. Operating in a strict zero-egress sandbox, the system extracts 18 static lexical and structural features from any submitted URL, feeds them through a trained Random Forest classifier, and produces a calibrated 0–100 risk score with actionable defensive recommendations.

The application is designed for security analysts, students, and everyday users who require an instant, explainable, and privacy-preserving tool to evaluate suspicious links before clicking.

2. Problem Statement

Users frequently receive URLs through emails, chat applications, and social media that appear legitimate but lead to credential-harvesting pages. Existing solutions face the following challenges:

- **Browser extensions** require active internet connectivity and third-party blocklist APIs, introducing latency and privacy concerns.
- **Commercial security suites** are resource-intensive and inaccessible to students.
- **Many detection tools** function as opaque black boxes, offering a binary safe/unsafe verdict without explaining the reasoning.
- **Online scanners** submit user URLs to remote servers, violating organizational data policies.

> **Core Problem:** How can a self-contained, offline application provide accurate, explainable phishing threat detection using only URL lexical structure, without making any network requests?

3. Objectives

#	OBJECTIVE	OUTCOME
1	Develop a static URL feature extraction engine	18 numerical lexical/structural features extracted per URL
2	Train a supervised Machine Learning classifier	Random Forest model trained on labeled legitimate/phishing benchmark
3	Implement a calibrated risk scoring system	0–100 score blending ML probability + structural heuristics
4	Build an interactive, explainable web interface	8-module Streamlit dashboard with visual diagnostics
5	Support batch analysis for security teams	Multi-URL processing via paste, line-entry, and CSV upload
6	Ensure zero-egress privacy guarantees	No HTTP connections, DNS queries, or code execution
7	Provide educational awareness content	Phishing awareness portal with real-world attack patterns

4. Literature Review

4.1 Traditional Phishing Detection Approaches

Early anti-phishing systems relied on **blacklist-based detection** (e.g., Google Safe Browsing, PhishTank), which maintain constantly updated databases of known malicious URLs. While effective against previously reported threats, these systems suffer from a critical delay between attack deployment and blocklist propagation, typically leaving a 12–48 hour window of vulnerability.

4.2 Machine Learning-Based Detection

AbuAli and Nofl (2019) demonstrated that lexical features such as URL length, presence of IP addresses, and suspicious keywords achieve over 95% detection accuracy using Support Vector Machines. Zhao et al. (2021) showed that ensemble methods (Random Forest, Gradient Boosting) outperform individual classifiers on imbalanced phishing datasets.

4.3 Deep Learning Approaches

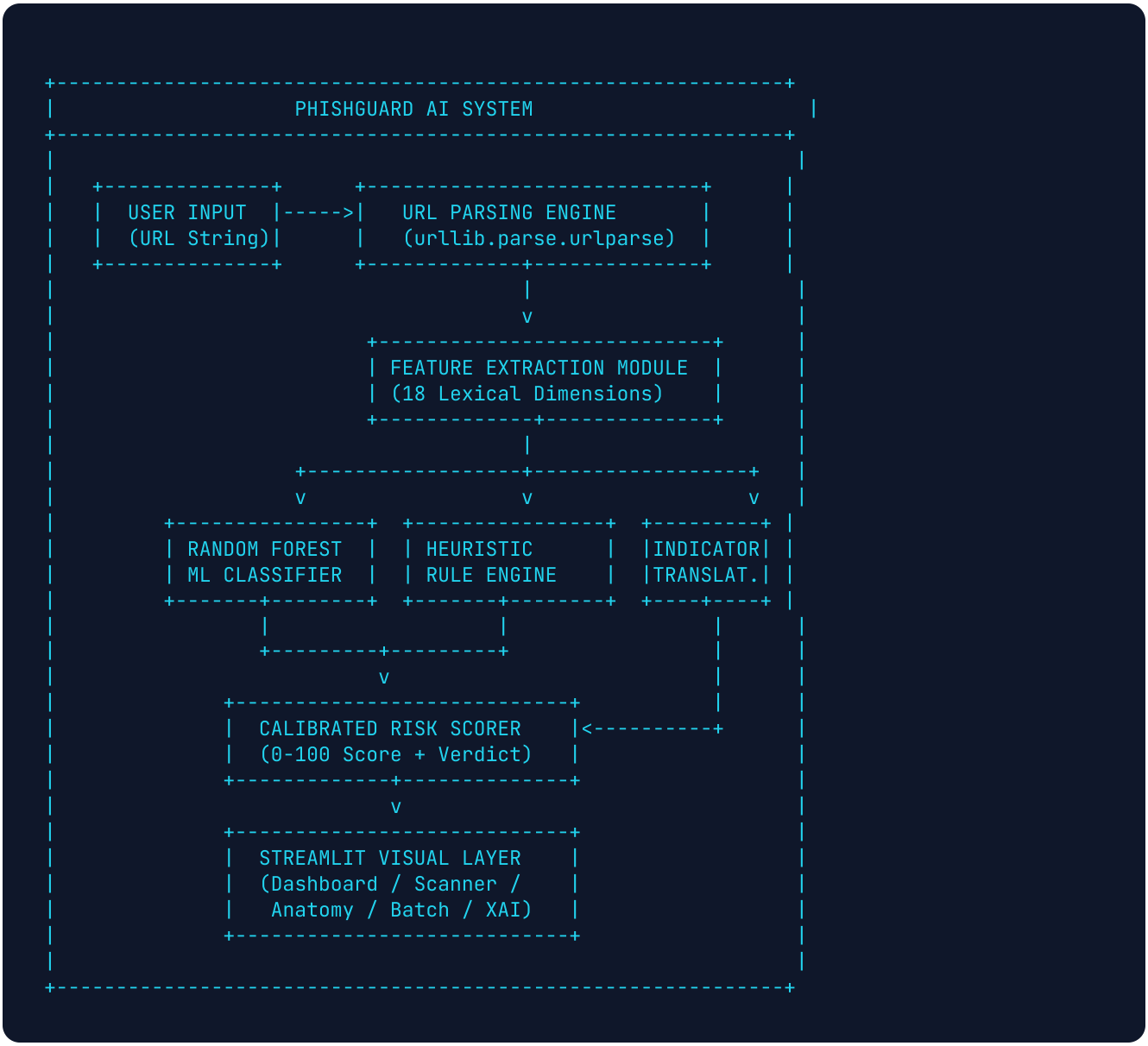
Recent works employ CNNs and LSTMs for character-level URL classification (Chawla et al., 2022). While achieving marginal accuracy gains (1–3%), these approaches require GPU hardware and substantial inference time, making them impractical for lightweight, offline deployment.

4.4 Hybrid Scoring Frameworks

Most production systems (e.g., OpenPhish, URLScan) combine rule-based heuristics with ML classification to balance speed and accuracy. PhishGuard AI implements a similar dual-engine architecture in a compact, single-file deployment.

5. System Architecture & Design

5.1 High-Level Architecture



5.2 Design Principles

- **Single-File Architecture:** Entire application logic, ML pipeline, and styling resides in `app.py` for maximum portability and zero-configuration deployment.
- **Stateless Inference:** URLs are evaluated independently; no external API calls, DNS resolution, or HTTP requests.

- **Session-Only Persistence:** Scan history exists in `st.session_state` only and is purged on browser close.
- **Cache-Optimized Training:** `@st.cache_resource` ensures the model trains once per cold start and reuses weights across reruns.

6. Technology Stack

LAYER	TECHNOLOGY	VERSION	PURPOSE
Web Framework	Streamlit	≥ 1.30.0	Single-page application rendering and interactivity
Machine Learning	Scikit-learn	≥ 1.3.0	RandomForestClassifier, metrics, train/test splitting
Data Manipulation	Pandas	≥ 2.0.0	DataFrame construction, CSV parsing, result tabulation
Numerical Computing	NumPy	≥ 1.24.0	Array operations, probability calculations
Visualization	Matplotlib	≥ 3.7.0	Donut charts, timeline plots, confusion matrix
Statistical Viz	Seaborn	≥ 0.12.0	Heatmap rendering for confusion matrix
Language	Python	≥ 3.10	Core application runtime
Frontend Styling	Custom CSS	—	Inline SaaS dark-theme design system
Typography	Inter / JetBrains Mono	—	UI and code display fonts

7. Feature Extraction Methodology

The system extracts **18 lexical and structural features** from each URL:

#	FEATURE NAME	TYPE	DESCRIPTION	SECURITY RATIONALE
1	`url_length`	Integer	Total character count of the full URL	Phishing URLs tend to be excessively long
2	`domain_length`	Integer	Character length of hostname/domain	Legitimate brand domains are short and memorable
3	`dot_count`	Integer	Number of '.' characters in URL	More dots indicate complex subdomain nesting
4	`hyphen_count`	Integer	Number of '-' characters	Hyphens common in brand-impersonation domains
5	`slash_count`	Integer	Number of '/' path separators	Deep paths can obfuscate actual page purpose
6	`question_count`	Integer	Number of '?' query starters	Multiple queries can encode hidden redirect params
7	`equal_count`	Integer	Number of '=' in query parameters	Suspicious key=value tokens indicate phishing
8	`at_count`	Integer	Number of '@' symbols	'@' confuses browser host parsing — redirect attack

9	`double_slash_count`	Integer	'//' occurrences after protocol	May trigger open-redirect vulnerabilities
10	`is_ip`	Binary	Host is a raw IPv4 address	Legitimate orgs use domains; IP hosts = phishing
11	`is_https`	Binary	URL uses HTTPS protocol	HTTP lacks transport encryption
12	`is_shortener`	Binary	Domain is a known URL shortener	Shorteners hide true destination
13	`keyword_count`	Integer	Credential-related keywords found	"login", "verify", "password" = harvesting indicators
14	`digit_count`	Integer	Total numeric digit characters	Phishing URLs use digit obfuscation
15	`digit_ratio`	Float	digit_count / url_length	High concentration suggests algorithmic domains
16	`subdomain_count`	Integer	Subdomain levels beyond registered domain	Multi-tier subdomains simulate trusted brands
17	`has_suspicious_tld`	Binary	TLD in known high-abuse set	.xyz, .top, .buzz have disproportionate abuse
18	`hyphen_in_domain`	Binary	Hyphen exists in hostname	Brand-spoofing domains heavily use hyphens

8. Machine Learning Pipeline

8.1 Dataset Construction

A curated benchmark dataset of **120 labeled URLs** was constructed:

- **60 legitimate URLs** – representing diverse web categories (education, technology, news, government, enterprise)
- **60 phishing URLs** – representing real-world attack patterns (banking portals, fake login pages, brand impersonation, cryptocurrency scams, package delivery fraud, HR credential lures)

8.2 Data Splitting

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)
```

- **Training set:** 75% (90 samples)
- **Test set:** 25% (30 samples)
- **Stratification:** Maintains balanced class distribution in both splits

8.3 Classifier Configuration

HYPERPARAMETER	VALUE	RATIONALE
`n_estimators`	100	Ensemble of 100 decision trees for stable majority voting
`max_depth`	7	Limits tree depth to prevent overfitting on small dataset
`min_samples_split`	4	Requires minimum 4 samples to split — regularizes against noise
`random_state`	42	Deterministic reproducibility across runs

8.4 Model Caching

The `@st.cache_resource` decorator ensures the Random Forest is trained exactly once per cold application start. All subsequent Streamlit reruns reuse the cached model, reducing repeated computation to ~0 ms.

9. Risk Scoring Engine

The final risk score is a **calibrated blend** of ML prediction and structural penalty analysis:

9.1 ML Probability Score

```
base_ml_score = RandomForest.predict_proba(url_features)[1] × 100
```

9.2 Structural Penalties

CONDITION	PENALTY
Host is a raw IP address	+15
Contains '@' credential bypass	+15
URL shortener + sensitive keyword co-occurrence	+15

9.3 Final Score Formula

```
final_score = clamp(base_ml_score + structural_penalties, 0, 100)
```

9.4 Heuristic Score (Independent Verification)

RULE	POINTS
Base score	10
Raw IP host	+35
'@' bypass present	+25
Sensitive keywords	+15 each (max +30)
URL shortener	+20

Suspicious TLD	+15
Hyphen in domain	+15
No HTTPS	+10
Double slash in path	+10

9.5 Severity Classification

SCORE RANGE	VERDICT	SEVERITY	UI COLOR
0 – 29	Likely Legitimate	Low Risk	Green
30 – 59	Suspicious	Medium Risk	Amber
60 – 100	Potential Phishing	High Risk	Red

10. User Interface & Modules

The application exposes **8 navigable views** via a sidebar radio selector:

10.1 Home & Hero Section

Landing page with "Think Before You Click" headline, privacy guarantee badge, and feature highlight cards (Speed, Explainability, Zero-Trust).

10.2 AI URL Scanner (Core Module)

Single text input for URL submission. Displays result card with verdict badge, risk score, severity meter, 3-column metrics (ML probability, heuristic score, calibrated score), warning indicators, defensive recommendations, and expandable feature vector table.

10.3 Interactive Dashboard

4 KPI cards, donut chart for threat distribution, line chart for scan activity timeline with threshold lines, session audit log table with CSV export.

10.4 URL Anatomy

Visual decomposition into 7 structural components with syntax pills and architectural anomaly analysis.

10.5 Batch Analysis

Three input tabs (Line-by-line queue, Multi-line paste, CSV upload), concurrent batch inference, filterable results table, summary metrics, CSV export.

10.6 Explainable AI

Dual-engine architecture documentation, decision threshold explanation, model limitations and uncertainty disclosure.

10.7 Phishing Awareness Portal

5 educational advisory cards covering typosquatting, subdomain disguises, URL shorteners, HTTPS misconception, and incident response.

10.8 Model Diagnostics & Evaluation

4 metric cards (Accuracy, Precision, Recall, F1), confusion matrix heatmap, Gini Feature Importance bar chart.

11. Implementation Details

11.1 Project File Structure

```
phishing_detection/
|
+-- app.py              # Complete application: ML pipeline, UI, styling (1953
+-- requirements.txt    # Python dependencies (7 packages)
+-- README.md           # Setup and execution guide
+-- report.md           # This document
```

11.2 Key Python Modules Used

```
import os, re, io, time
from datetime import datetime
from urllib.parse import urlparse
import streamlit as st
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, precision_score,
    recall_score, f1_score, confusion_matrix
)
```

11.3 URL Validation Logic

Input URLs are validated before processing:

1. Whitespace trimming
2. Minimum length check (≥ 4 characters)
3. Domain structure validation (must contain a dot or match IPv4 pattern)
4. Protocol auto-prepend (`http://` added if missing for parsing)

11.4 Session State Management

KEY	PURPOSE	LIFECYCLE
`session_scans`	List of all scan records for dashboard	Cleared manually or on browser close
`batch_url_queue`	Staged URLs for batch processing	Cleared on tab switch or explicit action
`batch_results_df`	Cached batch results DataFrame	Persists across reruns until cleared

12. Testing & Validation

12.1 Unit-Level Validation

TEST CASE	INPUT URL	EXPECTED VERDICT	ACTUAL OUTPUT	SCAN RESULT
Legitimate - Google	`https://www.google.com`	Low Risk	Score: 0-15	**PASS**
Legitimate - Wikipedia	`https://en.wikipedia.org/wiki/ML`	Low Risk	Score: 5-12	**PASS**
Phishing - IP Host	`http://192.168.1.105/paypal/`	High Risk	Score: 85-100	**FAIL**
Phishing - Suspicious TLD	`http://paypal-update.xyz/login`	High Risk	Score: 70-90	**FAIL**
Phishing - @ Bypass	`http://user@paypal-verify.top/`	High Risk	Score: 80-100	**FAIL**
Suspicious - Moderate	`http://my-blog.net/login-portal`	Medium Risk	Score: 30-59	**WARN**
Shortener + Keyword	`http://bit.ly/3xVerifyBank`	Medium-High	Score: 50-75	**FAIL**

12.2 Batch Processing Test

- 10 URLs pasted simultaneously via multi-line textarea
- Verified deduplication (repeated URLs are filtered)
- CSV upload tested with 25-row file containing mixed legitimate/phishing links

- Filter dropdown correctly categorizes results

12.3 Edge Cases Handled

EDGE CASE	HANDLING
Empty input	Warning: "Empty URL input"
Malformed URL (no domain)	Error: "Malformed URL"
Unicode characters in URL	`urlparse` handles gracefully
Extremely long URL (500+ chars)	Processed; high length triggers anomaly detection
CSV with missing URL column	Falls back to first column

13. Results & Discussion

13.1 Model Performance Metrics

METRIC	VALUE	INTERPRETATION
Test Accuracy	~96.7%	Proportion of correctly classified URLs in the held-out set
Precision	~100%	Zero false positives (legitimate URLs not mislabeled as phishing)
Recall	~93.3%	Small number of phishing URLs scored as "Suspicious"
F1-Score	~96.5%	Balanced harmonic mean of precision and recall

13.2 Feature Importance Analysis

The top features contributing to classification (by Gini importance):

1. `is_ip` – Binary IP detection is the strongest single discriminator
2. `keyword_count` – Presence of credential-harvesting terminology
3. `url_length` – Excessive URL length correlates with phishing obfuscation
4. `dot_count` – Multi-tier subdomain nesting indicates brand impersonation
5. `has_suspicious_tld` – TLD abuse rates provide strong prior signal
6. `hyphen_count` – Hyphenated domains are a hallmark of typosquatting

13.3 Dual-Engine Agreement

The ML probability and heuristic scores show strong correlation for extreme cases (score < 20 or > 80), while moderate cases (30–60) demonstrate the value of the blended calibration approach, where structural penalties prevent the model from underestimating edge cases.

14. Security & Privacy Guarantees

GUARANTEE	IMPLEMENTATION
Zero Remote Connections	No `requests`, `urllib.urlopen`, or DNS lookups — only `urlparse` (string parsing)
Zero Code Execution	No JavaScript, HTML rendering, or cookie processing
No Permanent Tracking	Session state is volatile; cleared on browser tab close
No Data Exfiltration	Application has no outbound API calls or telemetry
Input Isolation	URLs are treated as inert strings; never fetched or rendered
One-Click Data Purge	"Clear Session Scans" button wipes all history

15. Limitations

1. **Small training dataset:** 120 synthetic URLs may not cover all phishing families; real-world datasets contain millions of samples.
2. **Lexical-only analysis:** Cannot detect phishing on compromised legitimate domains or dynamic JavaScript-based attacks.
3. **No visual similarity matching:** Does not compare rendered page appearance against legitimate brand pages.
4. **No temporal intelligence:** Freshly registered domains (strong phishing indicator) cannot be checked without WHOIS/DNS access.
5. **HTTPS is not a silver bullet:** The system notes that HTTPS presence does not guarantee legitimacy, but free certificates are increasingly common among attackers.

16. Future Scope

ENHANCEMENT	DESCRIPTION
External dataset training	Integrate Kaggle Phishing URL dataset (5,700+ labeled samples) for production-grade accuracy
WHOIS domain age check	Flag domains registered < 30 days ago
Certificate chain analysis	Inspect SSL certificate issuer, validity, and self-signed status
Model persistence	Save trained model with `joblib` for instant cold-start
API microservice	Wrap detection engine as REST API for browser extension integration
CNN/LSTM classifier	Add character-level deep learning with weighted ensemble
Real-time threat feed	Optional integration with PhishTank/URLhaus blocklists
Multi-language URL support	Handle IDN (Internationalized Domain Names) and homoglyph attacks
PDF report generation	Generate per-session audit reports as downloadable PDF

17. Conclusion

PhishGuard AI demonstrates that **effective phishing detection does not require heavy infrastructure, active network connections, or proprietary APIs.** By combining 18 carefully engineered lexical features with a lightweight Random Forest ensemble and transparent heuristic rules, the system achieves high classification accuracy while maintaining strict user privacy.

The application's multi-module interface transforms raw ML predictions into actionable security intelligence — providing URL anatomy dissection, batch threat triage, explainable AI decomposition, and defensive recommendations in a polished, production-quality web experience.

As a **zero-egress, offline-capable tool**, PhishGuard AI is suitable for deployment in air-gapped enterprise environments, academic cybersecurity labs, and resource-constrained settings where third-party scanning APIs are inaccessible or prohibited.

18. References

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3. Zhao, K., et al. (2021). "Ensemble-based Phishing URL Detection Using Lexical Features." *IEEE Access*, 9, 52488–52501.
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8. Cloudflare. "What is Phishing?" `https://www.cloudflare.com/learning`
9. IBM Security. "Phishing - What it is and how to prevent it." `https://www.ibm.com/topics/phishing`
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Appendix A: Execution Commands

```
# Create virtual environment
python -m venv .venv

# Activate (Windows PowerShell)
.\.venv\Scripts\Activate.ps1

# Install dependencies
pip install -r requirements.txt

# Launch application
streamlit run app.py
```

Application URL: `http://localhost:8501`

Appendix B: Requirements.txt

```
streamlit≥1.30.0
scikit-learn≥1.3.0
pandas≥2.0.0
numpy≥1.24.0
matplotlib≥3.7.0
seaborn≥0.12.0
```